AGRIEDGE OR-MANGE LTD

Revolutionizing Agriculture Through Technology

Order Management System

Salesforce-Driven Solution

A Comprehensive Project Documentation

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Executive Summary

AgriEdge Or-Mange Ltd is undertaking a strategic digital transformation initiative to modernize its agricultural order management operations through a purpose-built Salesforce CRM solution. This document outlines the complete technical blueprint, system architecture, and implementation roadmap for the Order Management System (OMS).

The proposed solution addresses four core business imperatives:

- Process Automation — Eliminating manual workflows and reducing order processing time by automating task creation, notifications, and inventory updates.
- Real-Time Inventory Visibility — Providing instant, accurate stock levels with automated low-stock alerts and supplier notifications.
- Customer Service Enhancement — Empowering service teams with a 360-degree view of order status, history, and case management.
- Data-Driven Decision Making — Delivering executive dashboards and operational reports to support strategic planning.

1. Project Objectives

The AgriEdge OMS initiative is driven by the following strategic objectives, each aligned to measurable business outcomes:

- Automate Order Processing — Reduce manual intervention from order placement through fulfillment.
- Real-Time Inventory Tracking — Maintain accurate stock data with automated deduction upon order confirmation.
- Improve Customer Service — Provide service agents with complete order visibility and case management capability.

- Ensure Data Security — Implement role-based access control and field-level security to protect sensitive commercial data.
- Enable Reporting & Analytics — Generate actionable insights through dashboards and scheduled reports.

2. Data Architecture & Object Model

The system is built on a carefully designed Salesforce data model comprising six core objects. This architecture ensures data integrity, scalability, and clear relational boundaries across all business entities.

2.1 Core Objects

Object	Salesforce Equivalent	Purpose
Customer	Account / Contact	Stores customer master data and contact records
Product	Product2	Catalogue of agricultural products available for order
Order	Order__c (Custom)	Header-level order record with status and totals
Order Line Item	OrderItem__c	Line-level detail: product, quantity, and price
Inventory	Inventory__c (Custom)	Stock levels, reorder thresholds, and audit trail
Supplier	Supplier__c (Optional)	Vendor data for procurement and reorder workflows

2.2 Object Relationships

- One Customer (Account) → Many Orders [1:N]
- One Order → Many Order Line Items [1:N]
- One Product → Many Inventory Records [1:N]
- One Order Line Item → One Product [N:1 Lookup]

3. Fields & Formula Configuration

The following custom fields are configured across the primary objects to capture all required business data.

3.1 Order Object Fields

Field Name	Type	Configuration
Order Status	Picklist	Values: New | Processing | Shipped | Delivered
Order Date	Date	Auto-populated on record creation
Total Amount	Formula (Currency)	SUM of all Order Line Item totals

3.2 Inventory Object Fields

- Stock Available — Number field tracking current unit count in warehouse.
- Reorder Level — Threshold number; triggers low-stock alert workflow when breached.
- Last Updated — DateTime field auto-stamped by Apex trigger on every inventory change.

3.3 Formula Fields

Line Item Total Amount:

```
Quantity__c * Unit_Price__c
```

Low Stock Alert (Inventory):

```
IF(Stock_Available__c < Reorder_Level__c, "⚠ Low Stock", "✓ OK")
```

4. Validation Rules

Validation rules enforce data integrity at the point of entry, preventing erroneous records from being committed to the database.

Rule	Formula	Error Message
Positive Quantity	*Quantity__c <= 0*	Quantity must be greater than zero.
Valid Order Date	*Order_Date__c > TODAY()*	Order date cannot be in the future.
Mandatory Product	*ISBLANK(Product__c)*	A product must be selected for each line item.

5. Automation Design

Salesforce Flow and Process Builder automation handles three key business triggers, ensuring seamless handoffs between teams without manual intervention.

5.1 On Order Creation

- Automatically create a Processing Task and assign to the responsible Sales User.
- Send an email notification to the customer confirming receipt of their order.
- Log order creation timestamp and assign a unique reference number.

5.2 On Order Status = Confirmed

- Trigger Apex-based inventory deduction for each Order Line Item.
- Update Order status to 'Processing' and set expected fulfillment date.
- Notify the Inventory Manager of confirmed orders requiring dispatch.

5.3 On Low Stock Detected

- Formula field evaluates Stock Available vs. Reorder Level on every save.
- Flow triggers an alert email to the Inventory Manager and Admin.
- Optional: Auto-create a Supplier Purchase Request record for replenishment.

6. Apex Development

Apex code is employed for complex transactional logic that exceeds the capability of declarative automation tools. All Apex must achieve a minimum of 75% test coverage as required by Salesforce deployment standards.

6.1 Inventory Update Trigger

```
trigger OrderInventoryTrigger on Order__c (after insert, after update) {
  if (Trigger.isAfter) {
    InventoryService.deductStock(Trigger.new);
  }
}
```

6.2 InventoryService Apex Class

```
public class InventoryService {
  public static void deductStock(List<Order__c> orders) {
    for (Order__c o : orders) {
      List<OrderItem__c> items =
        [SELECT Product__c, Quantity__c FROM OrderItem__c WHERE Order__c = :o.Id];
      for (OrderItem__c oi : items) {
        Inventory__c inv =
          [SELECT Stock_Available__c FROM Inventory__c
           WHERE Product__c = :oi.Product__c LIMIT 1];
        inv.Stock_Available__c -= oi.Quantity__c;
        update inv;
      }
    }
  }
}
```

6.3 Test Class (75%+ Coverage Required)

```
@isTest
public class InventoryServiceTest {
  @isTest static void testDeductStock() {
    // Setup test data
    Product2 prod = new Product2(Name = 'Test Crop', IsActive = true);
    insert prod;
    Inventory__c inv = new Inventory__c(
      Product__c = prod.Id, Stock_Available__c = 100, Reorder_Level__c = 20);
    insert inv;
    // Create Order and trigger
    Order__c order = new Order__c(Status__c = 'Confirmed');
    insert order;
    // Assert inventory was reduced
    Inventory__c updated = [SELECT Stock_Available__c FROM Inventory__c
                            WHERE Product__c = :prod.Id];
    System.assertEquals(90, updated.Stock_Available__c);
  }
}
```

7. Security Architecture

Security is implemented across three layers to ensure data confidentiality, integrity, and appropriate access control.

7.1 Profiles & Roles

Role / Profile	Access Level	Permitted Actions
System Admin	Full Access	All objects, settings, users, and configurations

Sales User	Restricted	Create/edit orders; view products; no pricing edit
Inventory Manager	Restricted	Inventory CRUD; view orders; no customer data
Customer Service	Read-Only	View orders and cases; create cases only

7.2 Field-Level Security

- Unit Price and Discount fields — hidden from Sales Users and Customer Service profiles.
- Supplier Cost fields — visible only to Admin and Inventory Manager.
- Customer financial history — restricted to Admin and Sales Manager roles.

7.3 Sharing Rules

- Orders are visible only to the assigned Sales User and their manager in the role hierarchy.
- Inventory records are shared with Inventory Manager and System Admin only.
- Customer records follow standard territory-based ownership and sharing rules.

8. Reports & Dashboards

8.1 Standard Reports

- Orders by Status — Matrix report showing volume and value across all status stages.
- Inventory Levels by Product — Summary report with current stock vs. reorder thresholds.
- Low Stock Items Alert — Filtered report showing all products below reorder level.
- Order Fulfillment Time — Trend report tracking average processing-to-delivery duration.

8.2 Executive Dashboard Components

- Total Orders This Month — KPI metric component with month-over-month comparison.
- Orders by Status (Funnel) — Visual pipeline view from New to Delivered.
- Stock Alerts Panel — Live count of low-stock items with drill-down to report.
- Revenue vs. Target — Bar chart comparing actual order value to monthly target.

9. Integration Overview

At project scope, the following integrations are simulated or implemented at a basic level:

- Email Notifications — Salesforce native email alerts via Flow for order confirmation and stock alerts.
- Case Management — Automated Case creation when a customer complaint is logged against an order.
- Supplier Alerts — Email notification to supplier contact when stock falls below reorder level (optional extension).

10. End-to-End Process Flow

The following sequence describes the complete order lifecycle within the AgriEdge OMS:

#	Stage	Description
1	Order Placement	Customer places order via Sales User; Order record created in Salesforce with Status = New.
2	Task Assignment	Automation triggers: Processing Task assigned to responsible Sales User; confirmation email sent to customer.
3	Inventory Check	System validates stock availability for each Order Line Item via Apex trigger.
4	Order Confirmation	Sales User confirms order; Status updates to Confirmed; inventory automatically deducted.
5	Dispatch & Shipping	Warehouse team updates Status to Shipped; customer notified via email automation.
6	Delivery Confirmation	Status updated to Delivered; case creation enabled for any post-delivery customer issues.
7	Reporting	All transactions feed into dashboards and reports for management review in real time.

11. Submission Checklist

The following deliverables must be included in the final project submission:

✓	Deliverable	Status
☐	Data Model Diagram (ERD)	Completed
☐	Objects & Fields Documentation	Completed
☐	Automation Flows / Process Builder Screenshots	Completed

☐	Apex Code — Trigger, Service Class, Test Class	Completed
☐	Security Configuration — Profiles, Roles, Sharing Rules	Completed
☐	Sample Reports & Dashboard Screenshots	Completed
☐	End-to-End Use Case Explanation	Completed

This document is confidential and intended solely for internal project use.